

ACTIVITY SHEET ANSWERS

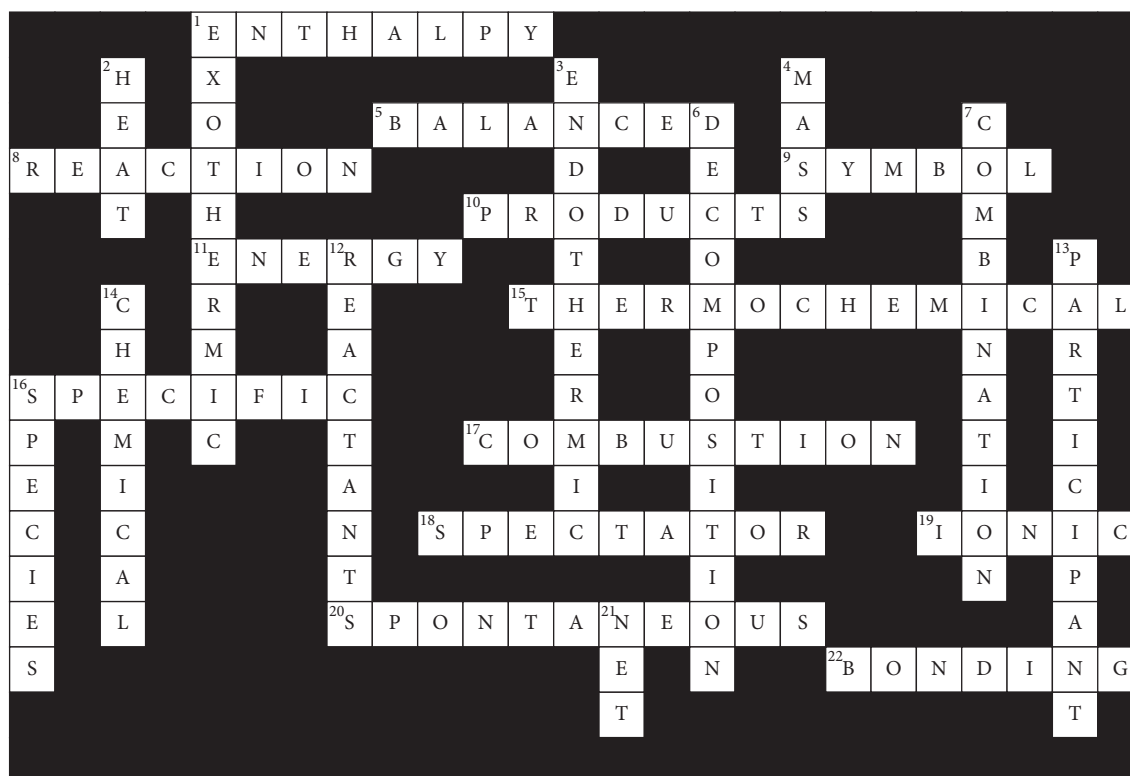
Chapter 4 Suggested answers

Prior knowledge

Beginning	What is another name for a chemical change?	Chemical reaction	What two changes are associated with chemical reactions?	Heat and light	What does a chemical reaction result in?
New substances	What is the scientific name for the new substances formed?	Products	What is the scientific name for the old substances that no longer exist?	Reactants	When two elements react together, what is the product called?
Compound	What kind of reaction forms a compound from two elements?	Combination reaction	What kind of reaction changes a compound into separate elements?	Decomposition reaction	What is the name given to the type of reaction that releases heat?
Exothermic reaction	What is the name given to the type of reaction that needs heat to keep it going?	Endothermic reaction	What happens to chemical bonds in a reaction?	Broken and reformed	What is required to break chemical bonds?
Energy	What happens to energy when bonds are formed?	Released as heat	What are reactions called that are written out in words?	Word equations	What does the arrow in a word equation mean?

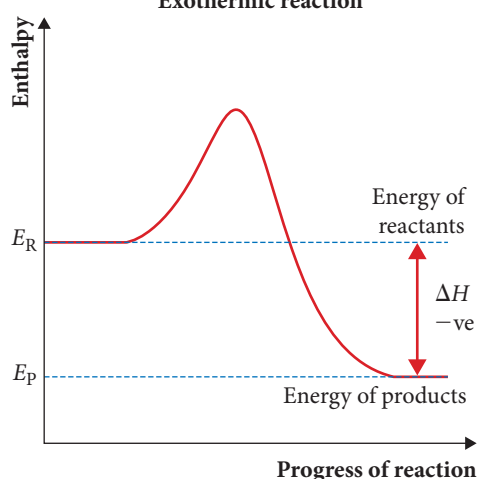
'Changed into'	What are reactions called that are written using chemical symbols?	Symbol equations	When is a symbol equation balanced?	When there are equal numbers of each type of atom on both sides of the equation	What principle requires equations to be balanced?
Conservation of mass	What type of reaction involves a rapid reaction with oxygen gas?	Combustion	What kind of biological reaction requires oxygen?	Respiration	What kind of biological reaction produces oxygen?
Photosynthesis	What type of chemical is combusted to release energy?	Fuel	Why would a combustion reaction be classified as exothermic?	Because it releases heat	Why would photosynthesis be classified as an endothermic reaction?
Because it requires light energy to sustain it	What gas product forms when metals react with acids?	Hydrogen gas (H₂)	What product forms when marble chips react with acid?	Carbon dioxide (CO₂)	What compound forms when hydrogen gas is combusted?
Water (H₂O)	What products are formed when a fuel is combusted in lots of oxygen?	Carbon dioxide gas (CO₂) and water (H₂O)	What factors can affect the rate of a reaction?	Concentration, catalysts and temperature	End

Activity sheet 4.1

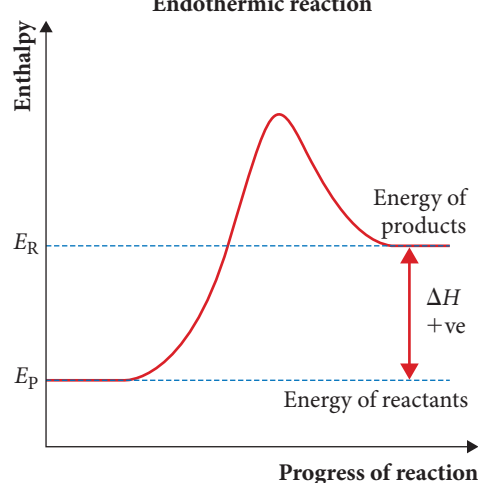


Activity sheet 4.3

1 Exothermic reaction



Endothermic reaction



- 2 The hump on each graph represents the energy required to get the reaction going. It is called the activation energy.
- 3 Because the graph shows that activation energy in the form of heat is required to start the reaction, after which the reaction will release more heat than it absorbs.
- 4 The graph would have no hump above the higher, dotted blue line; rather, it would descend from the energy of the reactants level down to the energy of the products level.

Activity sheet 4.4

- 1 Oxygen gas in the atmosphere.
- 2 Pure oxygen gas has many more oxygen molecules to collide with metal atoms than air has.
Reactions occur more rapidly at higher temperatures as particles collide with greater force.
- 3 Metal + oxygen $\rightarrow$ metal oxide
- 4 a $4\text{Li(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{Li}_2\text{O(s)}$
b $2\text{Ca(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{CaO(s)}$
c $4\text{Al(s)} + 3\text{O}_2\text{(g)} \rightarrow 2\text{Al}_2\text{O}_3\text{(s)}$
- 5 Metal + water $\rightarrow$ metal hydroxide + hydrogen gas
- 6 Reactions occur more rapidly at higher temperatures as particles collide with greater force.
- 7 a $2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH} + \text{H}_2\text{(g)}$
b $\text{Mg(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Mg(OH)}_2 + \text{H}_2\text{(g)}$
c Zinc does not react with water at room temperature.
- 8 metal + acid $\rightarrow$ salt + hydrogen gas
- 9 Because the reactions would be too violent. Because most metals react readily with dilute acids.
- 10 a $2\text{Li(s)} + 2\text{HCl} \rightarrow 2\text{LiCl} + \text{H}_2\text{(g)}$
b Copper does not react with dilute acids
c $\text{Mg(s)} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2\text{(g)}$
- 11 a $2\text{Na(s)} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{(g)}$
b $\text{Ca(s)} + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{H}_2\text{(g)}$
c $2\text{Al(s)} + 3\text{H}_2\text{SO}_4 \rightarrow \text{Al}_2(\text{SO}_4)_3 + 3\text{H}_2\text{(g)}$
- 12 $\text{K} > \text{Na} > \text{Li} > \text{Ca} > \text{Mg} > \text{Al} > \text{Zn} > \text{Fe} > \text{Pb} > \text{Cu} > \text{Ag} > \text{Au}$
- 13 Aluminium does not appear to react as strongly as it ought to with steam and acid.
- 14 It quickly forms a protective oxide coating, which prevents further reaction.

Activity sheet 4.5

- 1 To check whether the nail rusts in the absence of air when water, light and warmth are provided
- 2 By boiling the water to remove all the air and by covering the cooled boiled water with oil, which would prevent any further air from dissolving in the water
- 3 Oxygen gas
- 4 To check whether the nail rusts in the absence of moisture when air, light and warmth are provided
- 5 By inserting some desiccant in the test tube, then sealing it with a bung. The desiccant would absorb all of the moisture from the air.
- 6 To check whether the nail rusts in the absence of light when air, moisture and warmth are provided
- 7 By wrapping the test tube in foil, which would prevent any light from reaching the nail

- 8 To check whether the nail rusts in the absence of warmth when air, moisture and light are provided
- 9 By placing the test tube in the fridge and ensuring the internal light is kept on when the fridge door is closed
- 10 To check whether the nail would rust when all four factors are present
- 11 Usually all factors are present in a control, so test tube A would be considered a control.
- 12 That rusting will occur when all conditions are present
- 13 That rusting necessarily requires all four conditions
- 14 Air and moisture
- 15 Because the nail did not rust when air was absent in test tube B even though moisture, light and warmth were present; nor did it rust when moisture was absent in test tube C even though air, light and warmth were present
- 16 Light
- 17 Because rusting occurred in the complete absence of light
- 18 No, because temperature affects the rate of all reactions
- 19 Repeat the experiment but placing nails in a room, fridge and freezer and see whether there is a difference in the amount of rust produced.
- 20 Answers will vary.

Chapter 4 Revision

- 1 a $\rightarrow$ B, b $\rightarrow$ E, c $\rightarrow$ D, d $\rightarrow$ A, e $\rightarrow$ R, f $\rightarrow$ U, g $\rightarrow$ F, h $\rightarrow$ P, i $\rightarrow$ Q, j $\rightarrow$ X, k $\rightarrow$ J, l $\rightarrow$ W, m $\rightarrow$ L, n $\rightarrow$ I, o $\rightarrow$ G, p $\rightarrow$ N, q $\rightarrow$ O, r $\rightarrow$ T, s $\rightarrow$ S, t $\rightarrow$ K, u $\rightarrow$ M, v $\rightarrow$ V, w $\rightarrow$ C, x $\rightarrow$ H
- 2 a *Reactants* are the substances that are consumed in a chemical reaction. *Products* are the new substances formed in a chemical reaction.
b A *chemical equation* describes a chemical reaction using words. A *symbol equation* describes a reaction using chemical symbols.
c In a *combination reaction* atoms from two or more elements are combined into compounds. In a *decomposition reaction* a compound is broken down into simpler chemicals.
d In an *exothermic reaction* heat or light energy is released into the environment. In an *endothermic reaction* heat energy is absorbed from the environment.
e An *ionic equation* gives information about all of the species involved in reaction, particularly the ions. A *thermochemical equation* gives information about the change in enthalpy that occurs in a reaction.
f *Heat of combustion* is the amount of heat released when a fuel undergoes complete combustion at standard atmospheric pressure. *Specific heat capacity* is the amount of heat required to raise the temperature of 1 g of a substance by 1 degree kelvin.
- 3 a True
b True
c False (*each type of atom*)
d True

- e True
- f True
- g False (*absorbed* when bonds are broken and *released*)
- h True
- i False (*Net* ionic equations)
- j True
- k False (always *positive*)
- l False (*will rise by different amounts even*)

4

Reason why energy is required

to remove delocalised electrons from the metal atoms

to separate anions and cations in the lattice

to overcome electrostatic forces of attraction holding atoms together in molecules

- 5 a It is a balanced equation because the number of each type of atom is the same on both sides of the equation. There are 2 Mg and 2 O atoms on each side.
- b It is an exothermic reaction because ΔH is a negative value implying that heat is released.
- c It is not a spontaneous reaction because magnesium does not ignite at room temperature.
- 6 a It is a balanced equation because the number of each type of atom is the same on both sides of the equation. There are 2 Al, 6 H, 3 S and 12 O atoms on each side.
- b They are dissolved in water as the absence of any specified state after the formula of an ion implies they are in an aqueous state.
- c The spectator ions are the SO_4^{2-} ions because they are unchanged in the reaction.
- d $2\text{Al(s)} + 6\text{H}^+ \rightarrow 2\text{Al}^{3+} + 3\text{H}_2\text{(g)}$
- 7 a Heat energy is released in the reaction
- b Heat energy is absorbed in the reaction
- c The left-hand reaction because the products have less energy than the reactants.
- d The right-hand reaction because the products have more energy than the reactants.
- e No, because the hump shows that it requires some energy to get the reaction started.
- 8 a m for mass; C for specific heat capacity; ΔT for change in temperature
- b m in grams (g); C in $\text{J K}^{-1} \text{g}^{-1}$; ΔT in degrees kelvin (K)
- c joules (J)
- d Because a rise of 1 K is the same as a rise of 1°C .